

Cambridge International AS & A Level

CANDIDATE
NAME

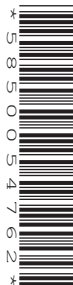
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CENTRE
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FURTHER MATHEMATICS

9231/41

Paper 4 Further Probability & Statistics

May/June 2020

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Blank pages are indicated.

	Grade awarded		
	A	B	C
Background music	49	51	40
Silence	93	68	49

Test at the 10% significance level whether grades awarded are independent of whether background music is playing during the examination. [6]

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6.44 6.16 5.62 5.82 6.51 6.62 6.19 6.42 6.34 6.28

It is claimed that the median time to complete the task is 6.4 milliseconds.

- (a) Carry out a Wilcoxon signed-rank test at the 5% significance level to test this claim. [6]

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- (b) State an underlying assumption that is made when using a Wilcoxon signed-rank test. [1]

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The random variable Y is such that $Y = \sqrt{X}$.

(b) Find the probability density function of Y .

[5]

[illegible]

The manager claims that there is no difference between the mean volume of coffee in cups filled by machine X and the mean volume of coffee in cups filled by machine Y .

[9]

[illegible]

- 5 A large number of children are competing in a throwing competition. The distances, in metres, thrown by a random sample of 8 children are as follows.

19.8 22.1 24.4 21.5 20.8 26.3 23.7 25.0

- (a) Assuming that distances are normally distributed, test, at the 5% significance level, whether the population mean distance thrown is more than 22.0 metres. [7]

[illegible]

- [illegible]

- [illegible]

[illegible]

The random variable Z is the sum of the number of red balls selected and the number of heads obtained by Rassa.

- (c) Find the probability generating function of Z , expressing your answer as a polynomial. [3]

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- (d) Use the probability generating function of Z to find $E(Z)$. [2]

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